

A Bilingual Latent Model of Proposition Hierarchy in Wittgenstein’s Tractatus

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Abstract

Wittgenstein’s Tractatus Logico-Philosophicus (1922) is suited to structural modelling because its 526 propositions encode a tree-like scaffold of hierarchy depth, local families, sequential order and child structure. This paper uses that scaffold as weak supervision for learning compact latent representations of proposition position, and asks whether such representations align German and English realisations of the same proposition. The monolingual study uses English texts; the bilingual study uses German and English versions, yielding 1052 language-specific samples aligned by proposition ID.

The retained model is a split-latent variational autoencoder (VAE) with gated recurrent unit (GRU) encoder and decoder. Its latent space separates a 24-dimensional text latent, used for reconstruction, from an 8-dimensional structure latent, used by structural heads predicting parent proposition, depth, next proposition and child count. In the bilingual model, the decoder is language-conditioned and an optional alignment loss pulls same-ID German and English structure latents together.

The monolingual seed sweep (10 seeds, regularisation at 0.05) shows that the split structure latent recovers hierarchy-relevant information. Parent accuracy is 0.6983 ± 0.0152 , depth accuracy 0.9274 ± 0.0275 and next-proposition accuracy 0.3724 ± 0.0198 , with reconstruction loss 1.2923 ± 0.0194 and perplexity 3.6416 ± 0.0706 . Sibling propositions are closer in structure space (2.0656 ± 0.0342) than unrelated propositions (5.5574 ± 0.0532).

The bilingual sweep across ten seeds and five alignment strengths shows strong cross-language proposition tracking. With no alignment loss, Top-1 retrieval is 0.9348 ± 0.0115 and mean reciprocal rank (MRR) is 0.9650 ± 0.0066 . Light alignment ($\lambda = 0.03$) gives the strongest mean scores: Top-1 retrieval 0.9365 ± 0.0103 , MRR 0.9662 ± 0.0058 , parent accuracy 0.8608 ± 0.0094 , depth accuracy 0.9643 ± 0.0136 and next accuracy 0.6189 ± 0.0219 . High alignment ($\lambda = 1.00$) degrades retrieval (0.7943 ± 0.0188) and MRR (0.8577 ± 0.0130). Shared hierarchy supervision therefore induces a compact proposition-level structure latent that aligns bilingual realisations without demonstrating general language-invariant logic or philosophical understanding of Wittgenstein’s text.

Keywords: digital humanities; Tractatus Logico-Philosophicus; variational autoencoder; representation learning; textual hierarchy; bilingual alignment; computational philosophy; latent structure.

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1 Introduction

Wittgenstein’s *Tractatus Logico-Philosophicus* is unusually suited to computational structural analysis because its propositions are not arranged merely as a linear sequence of philosophical remarks. They are organised by an explicit decimal numbering system that encodes relations of hierarchy and dependence.^{1,2} However, because the relation between tree-structured and sequential readings remains interpretively contested, the present paper treats the numbering system as a formal scaffold rather than as a complete philosophical interpretation.³ A proposition such as 4.121 is therefore not only a sentence in the text, but also a formal position within the work’s architecture. The numbering system indicates depth, parent propositions, local proposition families, sequential order, and child structure. This makes the *Tractatus* a rare philosophical corpus in which formal textual organisation can be operationalised directly.

Philosophical texts are usually studied in terms of argument, meaning, interpretation, and historical context. Yet some texts also possess explicit formal architectures that shape how their arguments unfold. Hierarchical and ordered documents occur not only in philosophy, but also in legal, theological, mathematical, and technical writing. Such texts raise a computational question: can a neural representation learn not only the wording of textual units, but also their position within an explicit document structure? The *Tractatus* provides a constrained and well-defined test case for this question because its proposition numbers supply a formal scaffold against which learned representations can be evaluated. Earlier computational work has assembled multilingual versions of the *Tractatus* as a parallel corpus and visualised proposition-level relations; the present study instead uses the numbering scaffold as weak supervision for latent representation learning.⁴

This paper asks whether the numbering scaffold of the *Tractatus* can support learning a compact latent representation of proposition position. The aim is not to show that a model understands Wittgenstein’s philosophy, nor that it discovers logical form from raw text alone. The claim is narrower: hierarchy-derived targets from the proposition numbering system can be used to train and evaluate a representation of formal textual position. In this sense, the task is supervised representation learning over a formally structured philosophical text, rather than unsupervised philosophical interpretation.

To test this, the study uses a split-latent variational autoencoder with recurrent encoder and decoder components, drawing on variational autoencoding and recurrent encoder-decoder modelling.^{5,6} The latent space is divided into a 24-dimensional text latent and an 8-dimensional structure latent. The text latent is used for reconstructing proposition wording, while the structure latent is trained to predict hierarchy-derived targets, including parent proposition, hierarchy depth, next proposition, and child count. This design encourages a division of labour between surface reconstruction and structural prediction. The purpose of the split is to make the hierarchy-carrying component of the model measurable and interpretable, rather than allowing all information to be absorbed into a single undifferentiated latent representation.

The first experiment is monolingual and uses English proposition texts. It tests whether the compact structure latent can recover hierarchy-relevant information from the English version of the *Tractatus*. The relevant question is not whether every formal relation is predicted with equal accuracy, but whether the learned representation captures broad hierarchy, local parenthood, and proposition neighbourhoods. This experiment establishes whether the model learns useful single-language structural information, and whether the hierarchy-oriented component behaves differently from a reconstruction-oriented text latent.

The second experiment extends the same framework to a bilingual setting. German and English versions of each proposition are aligned by proposition ID, producing two textual realisations of the same numbered position. The bilingual model uses shared hierarchy-derived targets, a shared

encoder, a shared bilingual vocabulary, and a language-conditioned decoder. It also includes an optional alignment loss that can pull German and English structure latents for the same proposition closer together. This setup makes it possible to test whether the shared targets can induce proposition-level alignment across languages without requiring open-ended translation modelling or pretrained multilingual embeddings.

The bilingual question is important because it separates surface language from document position. If German and English versions of the same numbered proposition occupy nearby regions of the structure latent space, then the model is not merely encoding language-specific wording. Cross-language retrieval of proposition IDs should nevertheless be interpreted carefully: it is evidence for constrained bilingual alignment under shared structural targets, not for general multilingual reasoning or philosophical understanding.

The results show that the monolingual structure latent captures hierarchy-relevant information, with depth recovered most strongly, parent prediction meaningful, and next-proposition prediction weaker. The bilingual results show strong German-English proposition retrieval. Importantly, explicit language alignment is not monotonically beneficial: low alignment gives the strongest overall performance, while stronger pressure degrades both hierarchy prediction and cross-language retrieval. This suggests that the shared space must balance cross-language invariance with fine-grained proposition discrimination.

The contribution of this paper is therefore methodological. It shows that the Tractatus numbering system can be operationalised as weak supervision for interpretable neural representation learning. It demonstrates that a split-latent VAE can partially separate reconstruction-oriented textual information from hierarchy-oriented structural information, and that bilingual versions of the same proposition can align when tied by common hierarchy-derived targets. The broader implication is that formally organised (e.g. philosophical, mathematical and legal) texts can serve as constrained test cases for studying how textual hierarchy, latent representation, and bilingual alignment interact.

The remainder of the paper describes the corpus, architecture, alignment objective, training procedure, and evaluation metrics. It then reports monolingual hierarchy recovery, bilingual proposition retrieval, reconstruction diagnostics, latent-distance analyses, and PCA visualisations before discussing the scope and limits of the findings.

2 Method

2.1 Experimental Design

The experiments comprise two parts of the same representation-learning study. The monolingual experiment tests whether a compact structure latent can recover hierarchy-derived targets from English proposition text. The bilingual experiment tests whether German and English versions of the same numbered proposition align while preserving the same hierarchy. The task is supervised representation learning over an explicitly structured philosophical text, not unsupervised discovery of meaning.

2.2 Corpus Construction and Hierarchy-Derived Targets

The corpus was constructed from the Project Gutenberg electronic text of the Tractatus, which provides the machine-readable source text used for proposition extraction.⁷ Each proposition is associated with a proposition ID, such as 4.121, which encodes its formal position in the book’s

hierarchy. The retained dataset contains all 526 propositions in the English version and, correspondingly, 1052 German and English propositions in the bilingual version. In the monolingual experiment, each row contains a proposition ID, English proposition text, and hierarchy-derived targets. In the bilingual experiment, each row contains the proposition ID, German text, English text, and shared hierarchy-derived targets. German and English versions are aligned by proposition ID, so the bilingual setup evaluates proposition-level structure alignment rather than unsupervised translation.

Hierarchy-derived targets are obtained from the proposition numbering system. The parent proposition is the immediately superordinate proposition in the numbering hierarchy. Depth records the hierarchical level of the proposition. The next-proposition target identifies the following proposition in reading order, and child count records the number of direct subordinate propositions. German and English versions of the same proposition share the same proposition ID, parent, depth, next-proposition target, child count, children, siblings, and ancestor chain.

A simplified bilingual row has this form:

```
id: "4.121"
parent_id: "4.12"
depth: 4
texts.en: "Propositions cannot represent the logical form..."
texts.de: "Die Saetze koennen die logische Form nicht darstellen..."
next_id: "4.1211"
children: ["4.1211", "4.1212"]
siblings: ["4.122", "4.123"]
ancestor_chain: ["4", "4.1", "4.12"]
child_count: 2
```

2.3 Tokenisation and Text Representation

Proposition text is tokenised at word level after lower-casing and punctuation handling. A vocabulary is constructed from the retained corpus and tokens are mapped to integer IDs. A padding token is used to support mini-batch training with variable-length proposition texts, and an unknown-token marker is used for tokens not present in the vocabulary. Token IDs are passed through a learned embedding layer. In the bilingual model, German and English tokens share the same token embedding table. This token embedding layer is not a pretrained multilingual embedding; it learns corpus-specific representations for both German and English vocabulary items during training.

2.4 Split-Latent VAE Architecture

The model is a recurrent variational autoencoder with a split latent representation. The encoder and decoder are implemented with gated recurrent units, following recurrent encoder-decoder sequence modelling.⁶ The encoder reads one proposition text at a time and maps the input sequence into latent parameters. The sampled representation is divided into a 24-dimensional text latent and an 8-dimensional structure latent. The text latent is used for reconstruction, whereas the structure latent is used for hierarchy prediction. This split makes the hierarchy-carrying component more interpretable.

The recurrent encoder receives tokenised proposition text after tokens have been embedded through the learned token embedding layer. The final recurrent state is projected into separate means and log-variances for the text and structure latents. Latent samples are drawn using the VAE reparameterisation trick, allowing signals to backpropagate through the sampling step.⁵ The

encoder is shared across samples; in the bilingual model it processes both German and English proposition texts.

The decoder reconstructs the input proposition text from the text latent and does not receive the structure latent. This design discourages the structure latent from being used as a general reconstruction feature and reserves it for the structural prediction heads. In the monolingual model the decoder reconstructs only English text. In the bilingual model the decoder is language-conditioned: a per-language embedding indicates whether the decoder should reconstruct German or English. The text latent plus an English language embedding is used to reconstruct English, whereas the text latent plus a German language embedding is used to reconstruct German. At each time step, the recurrent decoder outputs logits over the shared vocabulary. Applying a softmax yields token probabilities, and predicted token IDs are mapped back to strings using the corpus vocabulary.

The structure latent is passed to task-specific prediction heads. These heads predict parent proposition, depth, next proposition, and child count, and operate on the structure latent rather than on the text latent. Parent prediction, next-proposition prediction, and depth prediction are treated as categorical classification tasks. Child-count prediction is included as an additional count-valued supervised structural target. This use of related heads over a shared latent representation is consistent with multitask learning, in which auxiliary targets shape a common representation.⁸

2.5 Bilingual Conditioning and Structure Alignment

The bilingual model extends the monolingual split-latent architecture by adding German and English input texts, a shared bilingual token vocabulary, a shared encoder for both languages, a language-conditioned decoder, shared hierarchy targets by proposition ID, and, where enabled, a cross-language structure alignment loss. German and English versions of the same proposition are treated as different textual realisations of the same numbered position. The hierarchy targets used by the structural prediction heads are identical for samples with the same proposition ID.

Each bilingual sample carries a language label, English or German, which is mapped to a learned language embedding. This language embedding is concatenated with, or otherwise supplied alongside, the text latent during decoding. The purpose of language conditioning is to tell the decoder which surface language to reconstruct, reducing pressure on the 8-dimensional structure latent to account for language identity. The intended division of labour is that token embeddings and the text latent support language-specific reconstruction, the language embedding specifies the reconstruction language, and the structure latent supports hierarchy prediction.

The bilingual model can also apply a cross-language structure alignment loss to German and English structure latents for the same proposition ID. This loss penalises distance between same-ID English and German structure vectors and is calculated as a mean squared error. The purpose of the term is to test whether explicit cross-language pressure improves proposition-level alignment. Alignment strength is controlled by the language-alignment weight λ . The sweep tests alignment strengths of 0.00, 0.03, 0.10, 0.30, and 1.00. The no-alignment condition tests whether the shared structural targets are sufficient.

2.6 Training Objectives and Experimental Settings

The monolingual experiment uses English proposition texts. The model receives English text and hierarchy-derived targets. The retained monolingual configuration uses text latent = 24, structure latent = 8, $\beta_{\text{text}} = 0.01$, and $\beta_{\text{structure}} = 0.05$. The monolingual experiment evaluates whether the 8-dimensional structure latent captures hierarchy-relevant information. Reported monolingual

metrics include parent accuracy, depth accuracy, next-proposition accuracy, reconstruction loss, perplexity, KL structure, sibling distance, parent-child distance, and unrelated-proposition distance.

The full bilingual objective includes reconstruction loss, a Kullback–Leibler (KL) penalty for the text latent, a KL penalty for the structure latent, parent prediction loss, depth prediction loss, next-proposition prediction loss, child-count loss, and, where enabled, the cross-language structure alignment loss. The text and structure KL terms are weighted separately. The retained bilingual sweep uses $\beta_{\text{text}} = 0.001$, $\beta_{\text{structure}}=0.05$, $\lambda_{\text{parent}}=0.2$, $\lambda_{\text{depth}}=0.1$, $\lambda_{\text{next}}=0.2$, and $\lambda_{\text{child}}=0.02$. The language-alignment weight is varied while the other settings are held fixed. The monolingual loss is the bilingual loss without the bilingual structure alignment loss.

2.7 Evaluation and Statistical Reporting

Reconstruction quality is measured using reconstruction loss and perplexity. Structural prediction is evaluated using parent accuracy, depth accuracy, next-proposition accuracy, and child-count performance where applicable. Cross-language alignment is evaluated using proposition retrieval. In German-to-English retrieval, a German structure latent is used as the query and English structure latents are ranked by distance. Retrieval is correct if the nearest English latent has the same proposition ID; English-to-German retrieval is the reverse direction. Top-1 retrieval measures whether the nearest cross-language neighbour is the same-ID proposition. Mean reciprocal rank (MRR) measures how highly the correct same-ID proposition appears in the ranked retrieval list. Directional retrieval is reported separately to check whether alignment is balanced across English and German. By-language parent accuracy is also reported to check whether one language dominates structural prediction.

The learned structure latent is analysed geometrically for diagnostic purposes. PCA projections are used as representative visual diagnostics and are not treated as aggregate evidence. Each PCA figure corresponds to a specific trained seed and alignment condition. Language-coloured PCA plots test whether English and German samples separate into isolated language clusters. Depth-coloured PCA plots test whether the structure latent reflects hierarchy depth. Quantitative retrieval and hierarchy metrics are treated as the main evidence, with PCA used only to support interpretation of the learned latent geometry.

Reported quantitative claims are based on seed sweeps rather than single runs. The monolingual retrieval model is evaluated across seeds 0-9. Similarly, the bilingual alignment sweep is evaluated across ten seeds 0-9 for each alignment weight. Results are reported as means with SDs where appropriate.

3 Results

The learned 8-dimensional structure latent captures hierarchy-relevant information and supports English–German proposition alignment. This is evidence for constrained structural representation learning, not for comprehension of Wittgenstein’s logic.

3.1 Monolingual Split-Latent Results

The monolingual experiment establishes that the split-latent architecture can recover hierarchy-relevant structure from English proposition text while retaining separate reconstruction capacity. Table 1 and Figure 1 report the results. Depth prediction is strongest, parent prediction is moderate-to-strong, and next-proposition prediction is weaker, consistent with the greater difficulty

of recovering fine-grained sequence. The distance metrics provide a direct geometric check: siblings are much closer than unrelated propositions in the retained monolingual structure space.

Metric	Value
Seeds	10 (0-9)
Parent accuracy	0.6983 ± 0.0152
Depth accuracy	0.9274 ± 0.0275
Next accuracy	0.3724 ± 0.0198
Reconstruction loss	1.2923 ± 0.0194
Perplexity	3.6416 ± 0.0706
KL structure	8.9653 ± 0.1127
Sibling distance	2.0656 ± 0.0342
Parent-child distance	6.1065 ± 0.1566
Unrelated distance	5.5574 ± 0.0532

Table 1. Monolingual seed-sweep summary for the balanced split-latent configuration across seeds 0–9. Values are mean $\pm$ SD where an SD is reported. Depth accuracy is high, parent accuracy is moderate, and next-proposition accuracy is weaker. Reconstruction loss and perplexity report text-decoding performance, while KL structure indicates use of the 8-dimensional structure latent. The distance metrics separately assess hierarchy-related geometry: sibling propositions are substantially closer than parent-child or unrelated propositions.

3.2 Bilingual Alignment Results

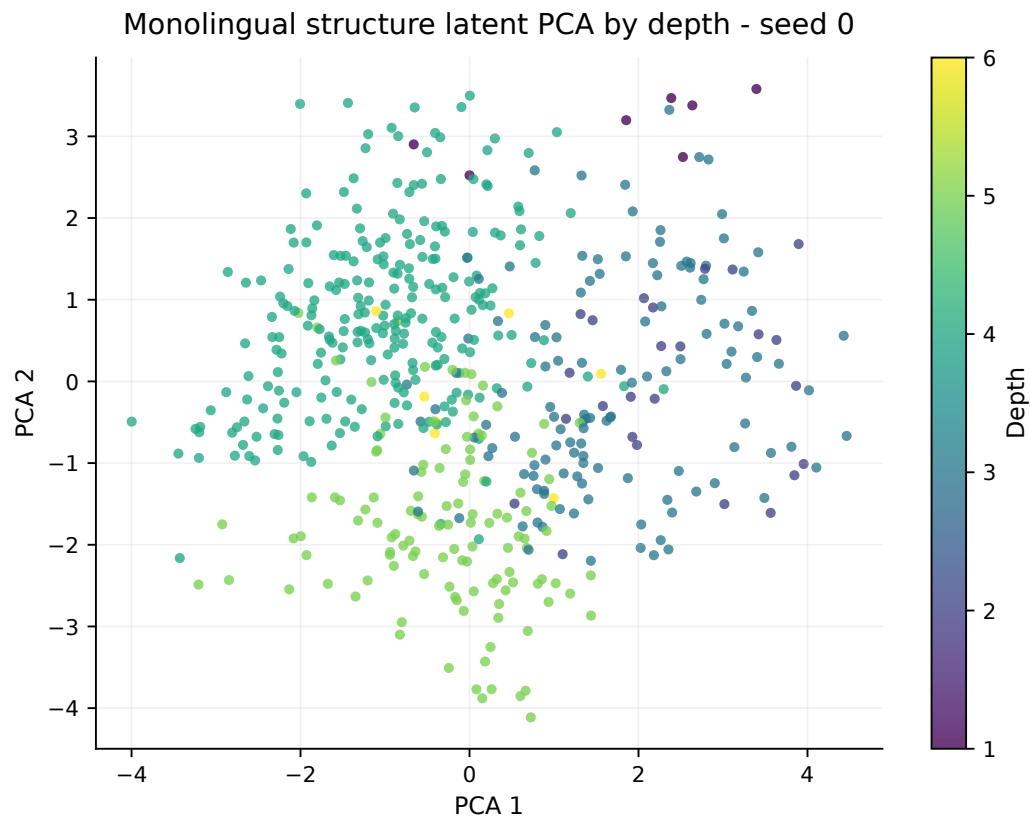
The bilingual experiment asks whether German and English versions of the same numbered proposition align while preserving the Tractatus hierarchy. The core seed-sweep results are reported in Table 2 and visualised in Figure 2. Explicit language alignment is not required for strong proposition retrieval: with $\lambda = 0.00$, Top-1 accuracy is 0.9348 ± 0.0115 and MRR is 0.9650 ± 0.0066 . Light alignment gives the strongest mean retrieval: at $\lambda = 0.03$, Top-1 accuracy rises to 0.9365 ± 0.0103 and MRR to 0.9662 ± 0.0058 . At $\lambda = 1.00$, Top-1 accuracy falls to 0.7943 ± 0.0188 and MRR to 0.8577 ± 0.0130 .

The same pattern is visible in the hierarchy-prediction metrics. As shown in Table 2 and Figure 3, parent, depth, and next-proposition accuracies are strongest under no or light alignment and deteriorate as λ increases. Depth is recovered most reliably, parent prediction remains strong in the low-alignment settings, and next-proposition prediction is the weakest structural task. The retrieval-structure trade-off is summarised in Figure 4: $\lambda = 0.30$ and especially $\lambda = 1.00$ reduce both cross-language retrieval and mean structure-prediction accuracy.

Directional retrieval was balanced across languages. Table 3 shows that German-to-English and English-to-German retrieval produce closely matched Top-1 and MRR scores at each alignment setting. This rules out a one-sided interpretation in which the model simply projects German into an English-centred space, or English into a German-centred space. Instead, both retrieval directions are supported by the same shared representation. By-language structure prediction shows the same pattern. As reported in Table 4, German and English parent, depth, and next-proposition accuracies are closely matched across the strongest settings, indicating that the model is not preferentially fitted to one language.

Reconstruction metrics provide useful diagnostics but are not sufficient for model selection. Table 5 and Figure 5 show that reconstruction loss and perplexity remain broadly stable under low alignment, but worsen at high alignment. Because the purpose of the model is not only to reconstruct proposition text but to isolate formal structure, reconstruction quality alone cannot

Figure 1. PCA projection of the 8D monolingual structure latent for representative seed 0, coloured by proposition depth. The figure is a single-seed diagnostic rather than an aggregate result, and should be read alongside Table 1.



identify the best model. The relevant model is the one that preserves both hierarchy prediction and cross-language proposition identity. The latent-distance diagnostics in Table 6 reinforce this interpretation: under low alignment, sibling propositions remain much closer than parent-child or unrelated propositions, whereas high alignment compresses the latent geometry and weakens structural separation.

The PCA projections of the learned structure latent support the quantitative results. Figure 6 shows the representative $\lambda = 0.03$, seed 0 structure latent coloured by language. German and English samples occupy the same broad latent space rather than separating into isolated language clusters. Figure 7 shows the same representative structure latent coloured by hierarchy depth. The visible depth organisation indicates that the shared bilingual structure latent retains hierarchy-relevant information while aligning German and English proposition samples. These PCA figures are diagnostic visualisations rather than aggregate evidence; the main claims rest on the ten-seed metrics reported in Tables 2–6.

Taken together, the bilingual results show that explicit structure in the Tractatus can be encoded in a compact latent space and shared across bilingual textual realisations. The result should be read narrowly: the model is not shown to understand Wittgenstein or to learn general language-invariant logic.

Lambda	Seeds	Parent accuracy	Depth accuracy	Next accuracy	Top-1	MRR
0.00	10 (0-9)	0.8601 $\pm$ 0.0094	0.9646 $\pm$ 0.0133	0.6202 $\pm$ 0.0194	0.9348 $\pm$ 0.0115	0.9650 $\pm$ 0.0066
0.03	10 (0-9)	0.8608 $\pm$ 0.0094	0.9643 $\pm$ 0.0136	0.6189 $\pm$ 0.0219	0.9365 $\pm$ 0.0103	0.9662 $\pm$ 0.0058
0.10	10 (0-9)	0.8539 $\pm$ 0.0102	0.9574 $\pm$ 0.0166	0.6142 $\pm$ 0.0237	0.9227 $\pm$ 0.0087	0.9546 $\pm$ 0.0043
0.30	10 (0-9)	0.8029 $\pm$ 0.0110	0.9320 $\pm$ 0.0249	0.5596 $\pm$ 0.0190	0.8720 $\pm$ 0.0112	0.9143 $\pm$ 0.0080
1.00	10 (0-9)	0.5736 $\pm$ 0.0340	0.7984 $\pm$ 0.0415	0.3614 $\pm$ 0.0213	0.7943 $\pm$ 0.0188	0.8577 $\pm$ 0.0130

Table 2. Core bilingual seed-sweep metrics across language-alignment weights. λ is the optional same-proposition German–English alignment weight. Parent, depth, and next accuracy evaluate the structural heads; Top-1 and MRR evaluate cross-language same-ID retrieval. All metric values are mean $\pm$ SD across seeds 0–9.

Lambda	GER to ENG Top-1	ENG to GER Top-1	GER to ENG MRR	ENG to GER MRR
0.00	0.9327 $\pm$ 0.0146	0.9369 $\pm$ 0.0109	0.9638 $\pm$ 0.0085	0.9662 $\pm$ 0.0060
0.03	0.9354 $\pm$ 0.0133	0.9376 $\pm$ 0.0100	0.9659 $\pm$ 0.0072	0.9666 $\pm$ 0.0054
0.10	0.9245 $\pm$ 0.0096	0.9209 $\pm$ 0.0089	0.9556 $\pm$ 0.0047	0.9536 $\pm$ 0.0047
0.30	0.8707 $\pm$ 0.0113	0.8732 $\pm$ 0.0133	0.9134 $\pm$ 0.0078	0.9151 $\pm$ 0.0095
1.00	0.7960 $\pm$ 0.0170	0.7926 $\pm$ 0.0229	0.8587 $\pm$ 0.0119	0.8566 $\pm$ 0.0149

Table 3. Directional cross-language retrieval performance across language-alignment weights. GER to ENG and ENG to GER report the two retrieval directions for Top-1 and MRR. All values are mean $\pm$ SD across seeds 0–9. Retrieval remains balanced, indicating reciprocal German–English proposition alignment rather than a one-way mapping.

Lambda	Parent GER	Parent ENG	Depth GER	Depth ENG	Next GER	Next ENG
0.00	0.8595 $\pm$ 0.0108	0.8606 $\pm$ 0.0091	0.9652 $\pm$ 0.0145	0.9641 $\pm$ 0.0131	0.6198 $\pm$ 0.0214	0.6205 $\pm$ 0.0210
0.03	0.8610 $\pm$ 0.0098	0.8606 $\pm$ 0.0101	0.9648 $\pm$ 0.0149	0.9637 $\pm$ 0.0128	0.6165 $\pm$ 0.0227	0.6213 $\pm$ 0.0246
0.10	0.8519 $\pm$ 0.0102	0.8559 $\pm$ 0.0122	0.9578 $\pm$ 0.0155	0.9570 $\pm$ 0.0182	0.6158 $\pm$ 0.0269	0.6125 $\pm$ 0.0224
0.30	0.8025 $\pm$ 0.0120	0.8034 $\pm$ 0.0124	0.9310 $\pm$ 0.0274	0.9331 $\pm$ 0.0233	0.5606 $\pm$ 0.0195	0.5586 $\pm$ 0.0201
1.00	0.5738 $\pm$ 0.0339	0.5734 $\pm$ 0.0355	0.8002 $\pm$ 0.0447	0.7966 $\pm$ 0.0399	0.3631 $\pm$ 0.0275	0.3597 $\pm$ 0.0164

Table 4. By-language structure-prediction performance across language-alignment weights. Parent, depth, and next accuracies are reported separately for German and English samples. All values are mean $\pm$ SD across seeds 0–9. The two languages remain closely matched, while high alignment weakens all three structural tasks.

Lambda	Reconstruction	Perplexity	KL text	KL structure	Same-ID distance
0.00	1.1893 $\pm$ 0.0193	3.2854 $\pm$ 0.0632	0.8843 $\pm$ 0.1962	10.0893 $\pm$ 0.1250	0.7598 $\pm$ 0.0195
0.03	1.1911 $\pm$ 0.0237	3.2914 $\pm$ 0.0778	0.9033 $\pm$ 0.1973	10.0786 $\pm$ 0.1272	0.7486 $\pm$ 0.0173
0.10	1.1880 $\pm$ 0.0215	3.2813 $\pm$ 0.0710	0.8596 $\pm$ 0.2033	9.9755 $\pm$ 0.1535	0.7756 $\pm$ 0.0140
0.30	1.2116 $\pm$ 0.0201	3.3594 $\pm$ 0.0673	0.7267 $\pm$ 0.1988	9.5007 $\pm$ 0.1895	0.9502 $\pm$ 0.0398
1.00	1.3198 $\pm$ 0.0290	3.7442 $\pm$ 0.1097	0.4379 $\pm$ 0.2191	7.3225 $\pm$ 0.4283	1.1970 $\pm$ 0.0398

Table 5. Reconstruction and latent-regularisation diagnostics across language-alignment weights. Reconstruction and perplexity measure token-level reconstruction; KL text and KL structure report latent use relative to the VAE prior; Same-ID distance reports the mean German–English structure-latent distance for the same proposition ID. All values are mean $\pm$ SD across seeds 0–9.

Lambda	Sibling	Parent child	Cross language parent child	Unrelated
0.00	1.8021 $\pm$ 0.0260	6.2126 $\pm$ 0.1088	6.2149 $\pm$ 0.1089	5.8365 $\pm$ 0.0649
0.03	1.8045 $\pm$ 0.0259	6.2038 $\pm$ 0.1104	6.2060 $\pm$ 0.1106	5.8314 $\pm$ 0.0663
0.10	1.8491 $\pm$ 0.0207	6.1420 $\pm$ 0.1215	6.1446 $\pm$ 0.1219	5.7964 $\pm$ 0.0741
0.30	2.0560 $\pm$ 0.0324	5.8920 $\pm$ 0.1310	5.8971 $\pm$ 0.1312	5.6303 $\pm$ 0.0797
1.00	2.4089 $\pm$ 0.0419	4.9136 $\pm$ 0.2078	4.9203 $\pm$ 0.2057	4.8078 $\pm$ 0.1599

Table 6. Structure-latent distance diagnostics across language-alignment weights. Sibling, parent child, cross-language parent child, and unrelated distances compare the geometry of selected proposition relations. All values are mean $\pm$ SD across seeds 0–9. Increasing λ , especially to 1.00, compresses the latent geometry and weakens structural separation.

Figure 2. Cross-language proposition retrieval across language-alignment weights. Points show means across seeds 0–9, with error bars indicating ± 1 SD. Top-1 measures nearest-neighbour same-ID retrieval in the opposite language; MRR measures the rank of the correct match. Retrieval is already strong at $\lambda = 0.00$, peaks at $\lambda = 0.03$, and declines under stronger alignment.

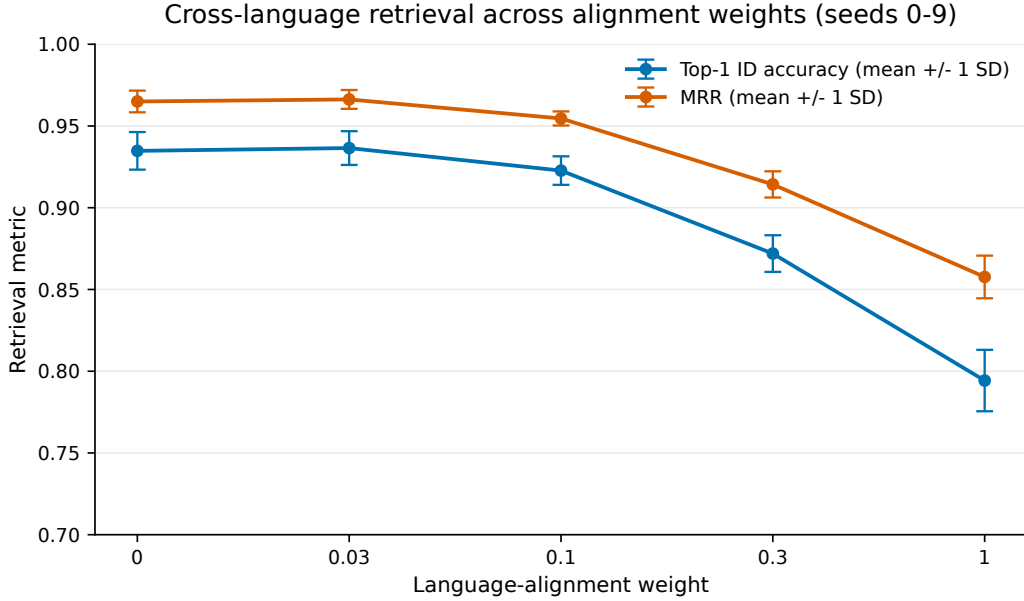


Figure 3. Structure-prediction accuracy across language-alignment weights. Points show means across seeds 0–9, with error bars indicating ± 1 SD. Depth is recovered most reliably, parent prediction remains strong at low alignment, and next-proposition prediction is the weakest structural task. Stronger alignment steadily degrades all three heads.

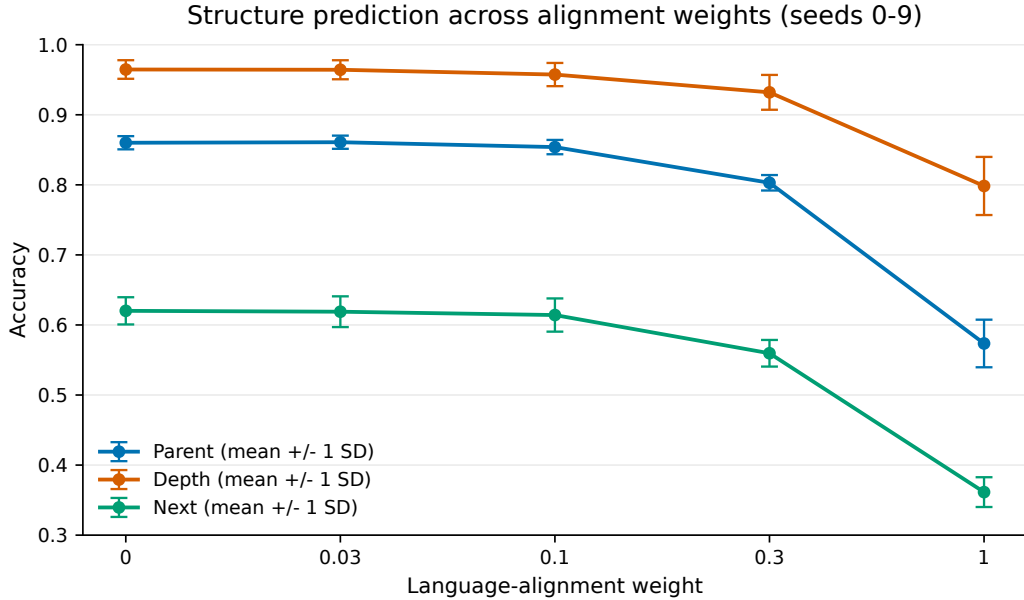


Figure 4. Retrieval-structure trade-off across language-alignment weights. Each point represents the ten-seed mean for a λ setting, with error bars showing ± 1 SD. The x-axis reports mean structure-prediction accuracy from the parent, depth, and next heads; the y-axis reports cross-language Top-1 proposition-ID retrieval. Low alignment occupies the upper-right region, whereas $\lambda = 1.00$ gives the weakest combined performance.

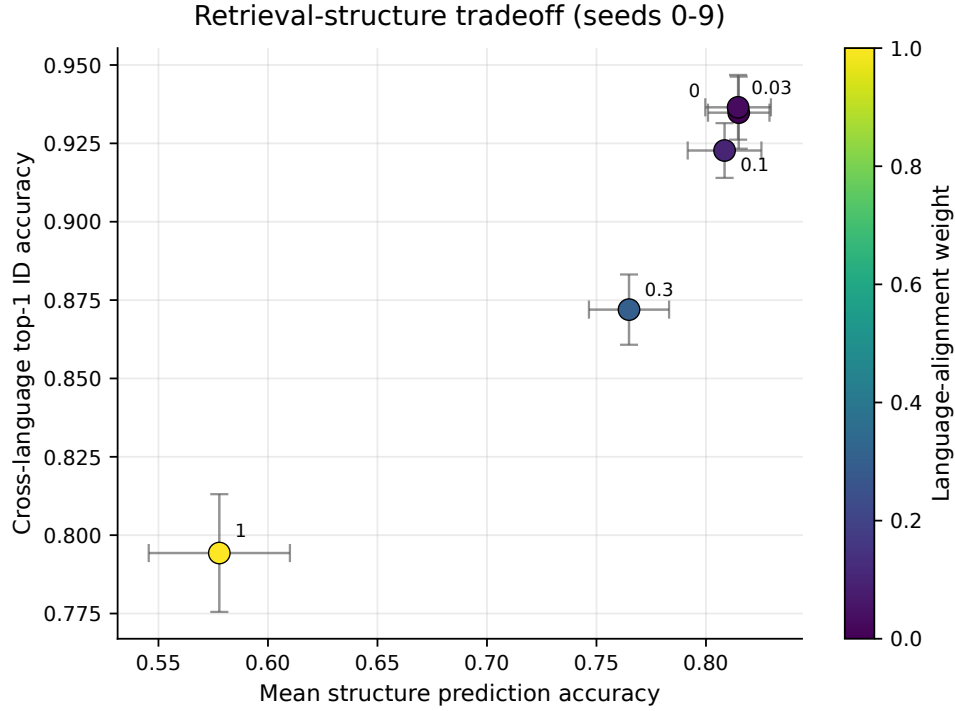


Figure 5. Reconstruction diagnostics across language-alignment weights. Points show means across seeds 0–9, with error bars indicating ± 1 SD. Reconstruction remains broadly stable at low alignment, but worsens as λ increases, especially at $\lambda = 1.00$. Reconstruction is therefore a useful diagnostic but not sufficient for model selection.

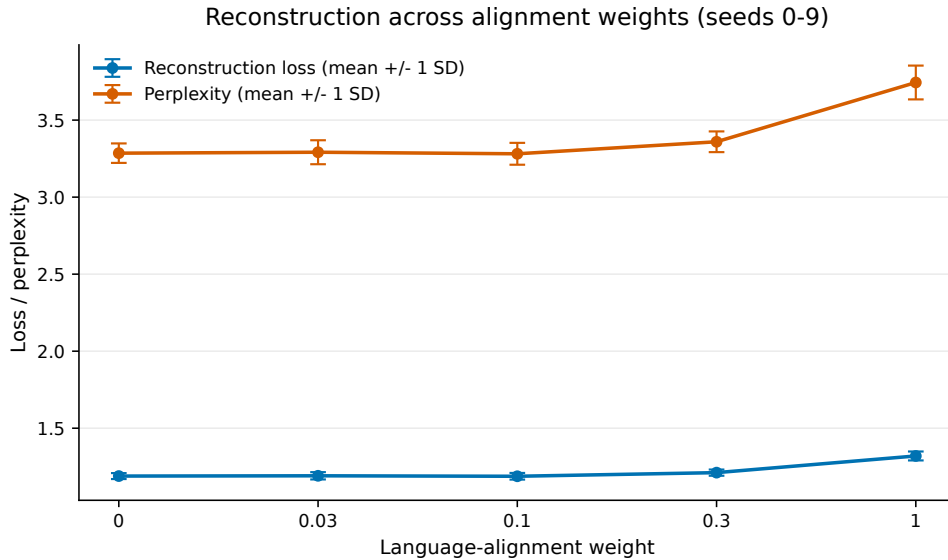


Figure 6. PCA projection of the bilingual 8-dimensional structure latent for representative $\lambda = 0.03$, seed 0, coloured by language. German and English samples occupy the same broad regions rather than separating into language-specific clusters. This is a single-seed diagnostic and should be read alongside the ten-seed retrieval and structure-prediction metrics.

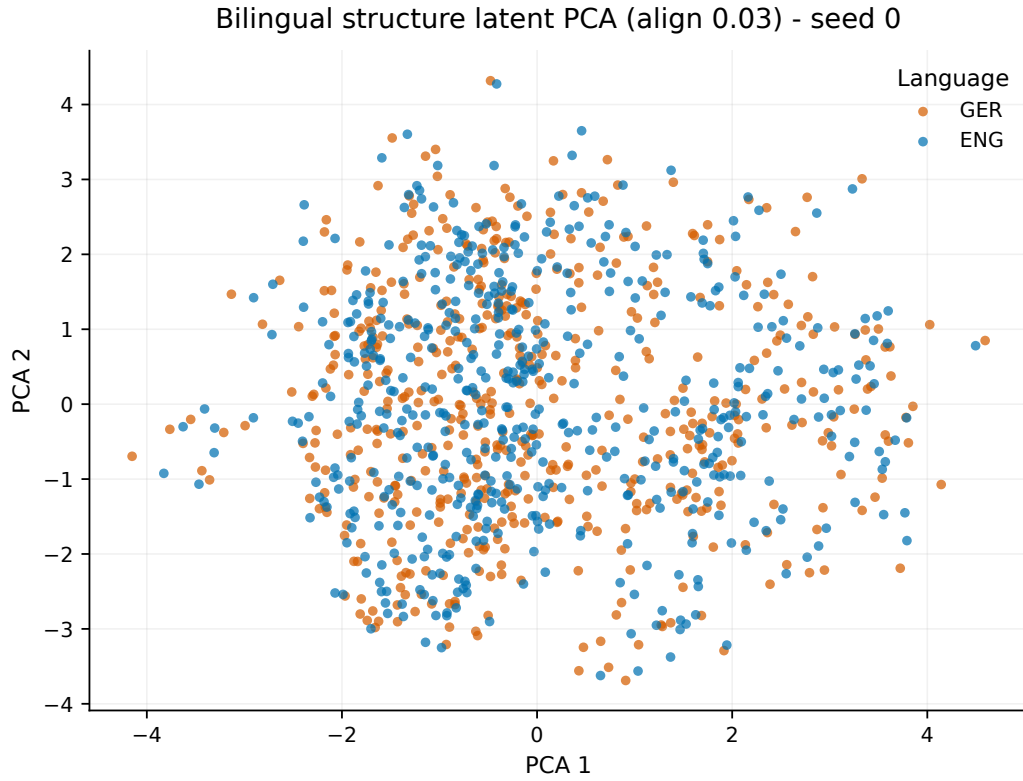
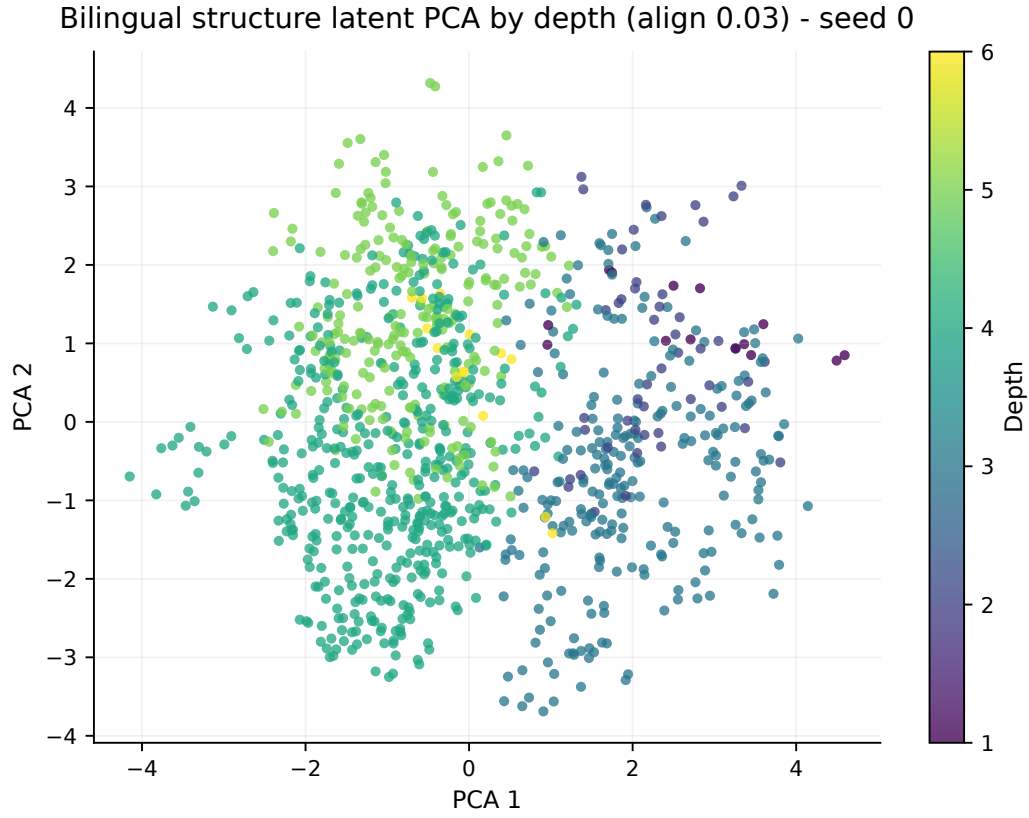


Figure 7. PCA projection of the bilingual 8-dimensional structure latent for representative $\lambda = 0.03$, seed 0, coloured by Tractatus hierarchy depth. The visible depth gradient indicates hierarchy-relevant organisation within the shared bilingual space. As with the other PCA figures, this is a single-seed visual diagnostic rather than aggregate evidence.



4 Discussion

The results support the view that the Tractatus numbering scaffold is not only an editorial device but also a usable source of structural supervision. By separating a reconstruction-oriented text latent from an 8-dimensional structure latent, the model makes it possible to ask whether information about hierarchy can be localised and evaluated. The evidence is strongest for broad hierarchical level and local parenthood, while next-proposition prediction remains weaker. This pattern is plausible: depth and parent relations are more stable properties of the numbering scaffold, whereas sequential order requires finer discrimination among neighbouring propositions.

The bilingual results extend this interpretation by showing that German and English versions of the same proposition can occupy a common structure space without requiring a translation objective or pretrained multilingual embeddings. Balanced German-to-English and English-to-German retrieval suggests that the space is not simply centred on one language. Instead, the aligned proposition IDs and shared hierarchy-derived targets provide a controlled bridge between different textual realisations of the same numbered position.

The alignment sweep is informative because it shows that cross-language invariance is not sufficient on its own: German and English versions of the same proposition should be close, but neighbouring propositions must remain distinguishable. Light alignment gave the best or near-best retrieval, whereas excessive pressure degraded both retrieval and hierarchy prediction. A plausible explanation is that too much alignment pressure may collapse neighbouring positions together, weakening the distinctions needed to preserve fine-grained hierarchy. The best model is therefore not the one that forces maximum German–English similarity, but the one that maintains enough latent separation to keep nearby propositions distinct.

The split-latent architecture is central to this interpretation. The decoder receives the text latent, while the structural heads act on the structure latent. This does not guarantee perfect disentanglement, but it creates a measurable hierarchy-carrying component and prevents reconstruction quality alone from defining success. The latent-distance diagnostics and PCA visualisations are useful in this respect: they do not replace the seed-sweep metrics, but they show that the learned space has interpretable geometric structure.

For digital humanities, the contribution is methodological. Formally organised philosophical texts can be treated as constrained corpora for evaluating representation learning over document structure. The Tractatus is a particularly clear case, but similar questions may arise in theological, legal, mathematical, and technical corpora where explicit ordering, hierarchy, or cross-lingual alignment is part of the textual object.

There are five key limitations. First, the hierarchy is derived from the numbering system rather than discovered from text alone; the model is therefore learning under weak supervision, and the latent factors should not be treated as unsupervised discoveries without inductive bias.⁹ Second, the Tractatus has an unusually explicit formal organisation, so transfer to less structured documents remains uncertain. Third, the latent structure should not be read as philosophical understanding: it captures formal position, not logical meaning. Fourth, the bilingual task uses aligned proposition IDs rather than open-ended translation or general multilingual reasoning. Fifth, the GRU encoder-decoder and the structural heads are deliberately modest, and next-proposition prediction remains comparatively weak. The PCA visualisations are also diagnostic, not definitive evidence.

Future work could test whether the learned structure latent generalises beyond seen proposition families, for example by holding back selected families and evaluating whether their structural positions can still be recovered. Stronger relational objectives for parent-child and sibling structure would also be useful. Graph neural networks are a natural extension because propositions could be represented as nodes and formal relations such as parenthood, siblinghood, sequence, and bilingual

equivalence as edges.¹⁰ A second extension would replace the GRU encoder with a Transformer-based encoder, which could provide richer text representations before projection into the text and structure latents.¹¹ In the present corpus, this would need careful testing under the same hold-out regime, since higher capacity could improve generalisation but could also overfit the small numbered corpus. Further examination of whether latent dimensions correspond to interpretable structural features may also be fruitful, as would testing with other hierarchical and formally organised texts.

5 Conclusion

This paper showed that the numbered structure of the *Tractatus Logico-Philosophicus* can supervise a compact latent representation of formal textual hierarchy. In the monolingual model, the structure latent recovered depth, parenthood, and local hierarchy. In the bilingual model, German and English versions of the same proposition aligned strongly in the shared structure space, provided that alignment pressure did not erase fine-grained distinctions. The result is a constrained computational model of textual hierarchy and bilingual latent alignment, not a claim of philosophical understanding. More broadly, the study treats a formally organised philosophical text as a constrained corpus for evaluating representation learning over document structure.

Data and code availability

Project code and derived experimental artefacts are available at:

[https://github.com/mark-alfred-griffiths/
Tractatus_Logico_Philosophicus_Bilingual_Study](https://github.com/mark-alfred-griffiths/Tractatus_Logico_Philosophicus_Bilingual_Study)

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